Risk-Aware Deep Reinforcement Learning for Collision Avoidance in Partially Observable Real-World Driving Scenarios

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Abstract

Some of the hardest collisions to avoid are those a driver cannot yet see coming: vehicles hidden behind blockers, pedestrians emerging from occluded crossings, conflicts developing outside a limited field of view. Most reinforcement learning (RL) results for driving assume the opposite – fully observed, synthetic traffic. We study risk-aware collision avoidance under partial observability, training deep RL agents on real-world driving scenarios from the Waymo Open Motion Dataset replayed in the Nocturne simulator, whose ray-cast visibility model restricts perception to what a physical sensor could see: hidden vehicles remain hidden. Our agent observes an ego-centric occupancy grid and analytic time-to-zero (TTZ) conflict features, both filtered to visible objects, together with *occlusion-risk summaries* that expose how much of the view cone is blocked and by what; a TTZ-barrier shaped reward penalizes risky closing behavior before a collision becomes unavoidable. A double dueling DQN is trained at scale with an asynchronous actor–learner architecture that runs unchanged from a single workstation to a Ray cluster. We additionally compare late-training dynamics for PPO and risk-aware R-MAPPO on the same occlusion-aware setup at episodes 400k–430k. In the completed DDQN held-out evaluation on 1,500 episodes, the agent reaches the goal in 95.33% of episodes (95% Wilson CI: 94.15–96.29%) with a 5.13% collision rate (95% Wilson CI: 4.13–6.37%). On the 1,089 episodes with non-degenerate trajectory traces, mean ADE is 2.24 ± 2.13 m, mean FDE is 6.40 ± 6.20 m, and mean absolute jerk is 10.27 ± 4.08 m/s^3.

1 Introduction

Autonomous vehicles must make safety-critical decisions with incomplete information. A parked truck hides a crossing pedestrian; a vehicle in the adjacent lane masks a merging car; a limited field of view leaves a fast-approaching agent undetected until seconds before conflict. In such *occluded* and *sensor-limited* situations the danger is by definition not yet visible when the avoiding maneuver must begin – which is precisely why occlusion is treated as a first-class hazard in the planning literature [5, 21]. Yet the majority of reinforcement learning (RL) results for driving are obtained under full observability in synthetic traffic, where every agent's state is available to the policy and the hardest part of the problem is assumed away.

This paper studies collision avoidance as a problem of decision-making under partial observability. We train and evaluate deep RL agents on real-world driving scenarios from the Waymo Open Motion Dataset [9] replayed in the Nocturne simulator [26], whose C++ ray-casting engine restricts the agent's perception to objects a physical sensor could actually see: vehicles behind blockers are invisible, and the view cone is limited in angle and range. The agent must reach a goal position extracted from the recorded human trajectory while avoiding collisions with road users that replay their logged behavior – confronting it with unprotected turns, merges, jaywalking pedestrians, and occluded conflicts drawn from real roads rather than procedurally generated traffic.

While imitation learning from human demonstrations produces strong driving policies [4, 6], it inherits the biases of the demonstration distribution and offers no mechanism for recovering from states human drivers rarely visit – precisely the states in which collisions occur. RL offers a complementary route, but introduces its own tension: a reward that pays for progress and penalizes collisions induces an explicit *safety–efficiency trade-off*, in which the agent may learn to accept a nonzero collision probability in exchange for reaching goals faster. Rather than treating this trade-off as a nuisance, we measure and characterize it directly.

Our contributions are as follows:

- **Occlusion-aware collision avoidance from real logs.** We formulate collision avoidance in replayed Waymo scenarios under a ray-cast visibility model, and design an observation that makes occlusion risk explicit: an ego-centric, class-weighted occupancy grid and analytic time-to-zero (TTZ) conflict features – both filtered to visible objects only – plus dedicated *occlusion-risk summaries* (occluded view fraction, nearest-blocker proximity, blocker density, and occlusion pressure) that allow the policy to modulate its behavior in response to unseen hazards (Section 4.1).
- **Risk-aware reward shaping.** We combine sparse terminal signals with a TTZ-barrier penalty that activates only when a conflict is imminent, providing a dense safety gradient before a collision becomes unavoidable (Section 4.2). The collision penalty R_c remains an explicit safety–efficiency knob; we report one operating point and leave a full Pareto sweep for follow-on work.
- **Stratified, statistically grounded evaluation.** We evaluate a trained DDQN checkpoint on 1,500 held-out episodes with Wilson intervals for binary rates, and stratify failures by vehicle TTZ imminence and episode length, alongside ADE/FDE and jerk-based smoothness (Sections 5–6). We further report late training returns for PPO and R-MAPPO as on-policy references.
- **A scalable training system.** A modern value-based learner (double dueling DQN with noisy exploration, n-step returns, prioritized replay, and the Muon optimizer [16]) is trained with an asynchronous actor–learner architecture in the style of Sample Factory [22] that scales from a laptop to a multi-node Ray cluster [20] without code changes (Sections 4.3–4.4).

On a held-out split of Waymo scenarios, the completed single-checkpoint evaluation covers 1,500 episodes and yields a 95.33% goal rate and 5.13% collision rate. The trajectory subset with valid multi-step traces yields 2.24 ± 2.13 m ADE, 6.40 ± 6.20 m FDE, and 10.27 ± 4.08 m/s^3 mean absolute jerk. Collision mass concentrates in low-TTZ episodes (15.3% collision when vehicle TTZ < 1 s versus 0% when $\mathrm{TTZ} \geq 3$ s).

The remainder of the paper is organized as follows. Section 2 positions our work relative to safe RL, occlusion-aware planning, and scalable RL systems. Section 3 formalizes the task as a POMDP, and Section 4 details the observation, reward, agent, and training system. Section 5 defines the metrics and statistical protocol; Section 6 presents the held-out DDQN evaluation, PPO/R-MAPPO training references, and stratified analyses.

2 Related Work

Driving simulators from real-world logs. CARLA [7] and SUMO [18] provide synthetic traffic, while Nocturne [26], Waymax, and nuPlan replay logged human trajectories, exposing agents to realistic interaction patterns. We adopt Nocturne for its fast C++ core, ray-cast partial observability, and native support for the Waymo Open Motion Dataset [9].

RL for collision avoidance. Value-based and policy-gradient methods have been applied to lane keeping, merging, and intersection handling [15, 17]. Occupancy-grid observations are a common mid-level representation that balances expressiveness and sample efficiency [11]. Time-to-collision-style features have long been used in classical safety systems [19]; our TTZ features bring this inductive bias into the observation and the reward.

Safety-critical and risk-aware RL. Safe RL formalizes the tension between task performance and constraint violation, via constrained MDPs [1, 3], risk-sensitive objectives that penalize variance or tail outcomes of the return [12], and shielding or barrier-style mechanisms that intervene near constraint boundaries [2]. Our TTZ-barrier penalty is a shaped, differentiable-in-effect analogue of a control-barrier condition: it applies increasing pressure as the most imminent conflict approaches, rather than penalizing only terminal collisions. Unlike hard-constraint methods we do not guarantee constraint satisfaction; instead we expose the safety–efficiency trade-off explicitly by reporting the measured operating point under a fixed penalty coefficient, following the position that in learned driving stacks the operating point should be a deliberate, measurable choice.

Driving under occlusion. Planning literature treats occlusion by reasoning over phantom agents or worst-case reachable sets behind blockers [21], and POMDP formulations marginalize over hypotheses about unobserved traffic [5]. These approaches hand-design the occlusion response. In contrast, we give a model-free agent explicit, low-dimensional

occlusion-risk statistics computed from ray-cast visibility and let it *learn* when caution is warranted, evaluating on real scenarios whose occlusion patterns arise from logged traffic rather than constructed worst cases.

Imitation learning baselines. Behavior cloning remains a strong supervised baseline for driving [4, 6], but suffers compounding errors under distribution shift [23] – exactly the regime of near-collision states. A matched BC held-out evaluation is left for follow-on work; this manuscript focuses on the completed DDQN evaluation and on-policy training references.

Value-based deep RL. Our agent composes established components: double Q-learning [25], dueling architectures [27], prioritized experience replay [24], noisy networks for exploration [10], and multi-step returns [13]. We additionally adopt the Muon optimizer [16], which orthogonalizes momentum updates for hidden weight matrices via Newton–Schulz iteration and has shown strong results in large-scale supervised training; we treat it as a drop-in optimizer choice and ablate it against AdamW rather than claiming it as a contribution.

Scalable RL systems. High-throughput RL architectures decouple acting from learning: Ape-X [14], IMPALA [8], and Sample Factory [22] demonstrate that asynchronous experience collection with centralized (batched) inference dramatically improves hardware utilization. Our system adapts this recipe to a CPU-heavy simulator with an off-policy learner, and layers Ray [20] on top for multi-node scaling.

3 Problem Formulation

We consider an episodic, partially observed Markov decision process $(\mathcal{S}, \mathcal{A}, P, r, \gamma, \Omega, O)$. Each episode instantiates one scenario from the Waymo Open Motion Dataset replayed in Nocturne at $\Delta t = 0.1\,\text{s}$. The state $s \in \mathcal{S}$ comprises the full simulator state – poses and velocities of all road users plus static map geometry. A single ego vehicle is controlled by the policy; all other road users – vehicles, pedestrians, and cyclists, the latter two referred to jointly as vulnerable road users (VRUs) – replay their logged trajectories, so the transition kernel P is deterministic given the ego action. The observation function O is *not* the identity: it first applies Nocturne's ray-cast visibility model, which removes every object outside the ego's view cone or occluded behind a nearer object, and then encodes the surviving objects into the feature vector $o \in \Omega \subset \mathbb{R}^{613}$ described in Section 4.1. Partial observability in our setting is therefore physical (sensor-like occlusion), not an artifact of feature compression. The ego receives the goal position $g \in \mathbb{R}^2$ recorded at the end of the human trajectory and must reach it within a tolerance while avoiding collisions with other objects and road edges. Episodes terminate on goal achievement, collision, or a horizon of $T_{\max} = 80$ steps (8 seconds).

The action space is a discrete product of throttle and steering levels, $\mathcal{A} = \{-1.0, 0.0, 2.0\} \times \{-0.7, -0.2, 0.0, 0.2, 0.7\}$, yielding $|\mathcal{A}| = 15$ actions executed through Nocturne's kinematic bicycle model.

4 Method

4.1 Observation Design

The observation is a flat vector concatenating four groups of features (Figure 1).

Ego-centric occupancy grid. We rasterize the local scene into a $C \times H \times W$ grid with $C = 3$ channels, $H = 20$ rows, and $W = 10$ columns, covering $40\,\text{m}$ ahead, $5\,\text{m}$ behind, and $\pm 10\,\text{m}$ laterally in the ego frame. Channel 0 encodes occupancy strength with class-dependent weights: vehicles ($w_{\text{veh}} = 1.0$), vulnerable road users (pedestrians and cyclists, $w_{\text{vru}} = 2.0$), and road edges ($w_{\text{edge}} = 3.0$); overlapping stamps take the maximum. Channels 1–2 encode the relative longitudinal and lateral velocity (normalized by $30\,\text{m/s}$) of the *nearest* vehicle occupying each cell, resolved by a vectorized per-cell minimum-distance reduction. Road-edge geometry is cached in the world frame once per scenario and re-projected into the ego frame each step.

Ego and goal features. Four ego-state features (heading, speed, heading error to goal, speed error to the logged target speed, each normalized) and two goal features (distance to goal normalized by $100\,\text{m}$; relative bearing normalized by π).

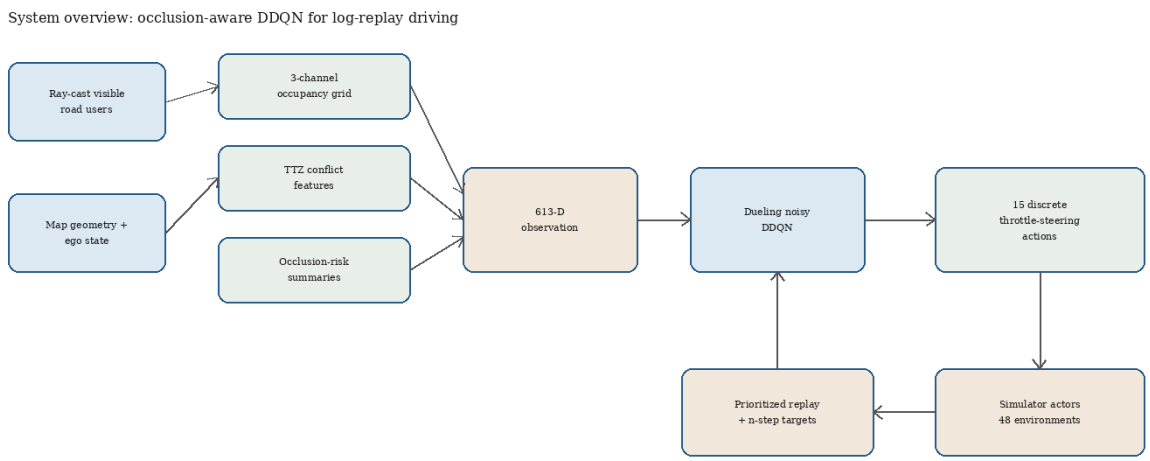


Figure 1: Overview of the implemented observation and learning pipeline. Ray-cast visible actors and map context form occupancy, TTZ, and occlusion-risk features; the 613-D observation is processed by a dueling noisy DDQN, and experience from 48 synchronous environments is replayed with prioritized sampling and multi-step targets.

Time-to-zero (TTZ) conflict features. For every non-ego object i at displacement $\mathbf{d}_i$ from the ego, we compute the scalar closing speed along the line of sight,

$$v_i^{\text{close}} = \frac{\mathbf{v}_{\text{ego}} \cdot \mathbf{d}_i}{\|\mathbf{d}_i\|} - \frac{\mathbf{v}_i \cdot \mathbf{d}_i}{\|\mathbf{d}_i\|}, \qquad \text{TTZ}_i = \begin{cases} \|\mathbf{d}_i\| \,/\, v_i^{\text{close}} & v_i^{\text{close}} > 0, \\ \text{TTZ}_\infty & \text{otherwise}, \end{cases} \tag{1}$$

where TTZ_∞ is a finite sentinel (999 s) and contact distances below 0.5 m map to $\text{TTZ} = 0$. The observation exposes the minimum TTZ over vehicles, the minimum over pedestrians and cyclists (both clipped to 20 s), and their difference – a compact, analytic summary of the most imminent conflict of each class.

Occlusion-risk features. A single call per step to Nocturne's C++ ray-casting engine returns the set of visible objects within the view cone. This visibility set serves two roles. First, it filters the occupancy grid and TTZ features above, so the agent only perceives objects a real sensor could see. Second, from the angular intervals subtended by visible blockers we compute four summary statistics that quantify what the agent *cannot* see: the occluded fraction of the view cone, the normalized distance to the nearest blocker, blocker density, and an "occlusion pressure" term (occluded fraction weighted by blocker proximity). These features are part of the full model evaluated in Section 6; an ablation that removes the four summaries while keeping visibility filtering is a natural follow-on experiment once a matched checkpoint is available.

The full observation has dimension $3 \times 20 \times 10 + 4 + 2 + 3 + 4 = 613$.

4.2 Reward

The per-step reward combines sparse terminal events with dense shaping:

$$r_t = \underbrace{R_g \mathbb{1}[\text{goal}]}_{\text{goal bonus}} - \underbrace{R_c \mathbb{1}[\text{collision}] - R_o \mathbb{1}[\text{road-edge collision}]}_{\text{safety penalties}} - \underbrace{c_s}_{\text{step cost}} + \underbrace{c_p\,(d_{t-1} - d_t)}_{\text{progress}} + \underbrace{c_h\Big(1 - \tfrac{|\theta_t|}{\pi}\Big)}_{\text{heading}} - \underbrace{c_z \max\big(0,\ \tau - \text{T}}_{\text{TTZ barrier}} \tag{2}$$

where d_t is the distance to goal, θ_t the heading error, $\text{TTZ}_t^{\min}$ the minimum TTZ over all object classes, and $\tau = 3$ s the safety threshold. We use $R_g = 100$, $R_c = 100$, $R_o = 50$, $c_s = 0.1$, $c_p = 2.0$, $c_h = 0.3$, $c_z = 0.5$. The shaping terms are active only on non-terminal steps. The TTZ barrier is the key safety-shaping component: it grows linearly as the most imminent conflict drops below τ, providing gradient signal *before* a collision occurs rather than only at the terminal event.

Algorithm 1 Asynchronous training loop (main process)

Spawn N_w actor workers (N_e envs each) with shared-memory buffers
Spawn learner process; wait for initial inference weights
while total steps $<$ budget **do**
  Collect ready observations from workers (shared memory)
  $a \leftarrow \arg\max_a Q_\theta(s, a)$ (single batched GPU forward)
  Dispatch actions to workers; receive (s, a, r, s', d) batches
  Submit transition batch to learner queue (bounded, overload policy)
  Poll learner: ingest metrics; load refreshed weights if published
end while

4.3 Agent

The agent is a double dueling DQN [25, 27]. The grid channels and auxiliary features are encoded and fused by an MLP trunk (4 hidden layers of width 256), followed by dueling value and advantage heads. All exploration is state-dependent via noisy linear layers [10], annealed jointly with a small ϵ-greedy floor ($\epsilon : 1.0 \to 0.05$ over 5×10^5 steps). Targets use n-step returns ($n = 3$) with double Q-learning bootstrapping:

$$y_t = \sum_{k=0}^{n-1} \gamma^k r_{t+k} + \gamma^n \, Q_{\bar{\theta}}\big(s_{t+n}, \, \arg\max_a Q_\theta(s_{t+n}, a)\big). \tag{3}$$

Transitions are sampled from a prioritized replay buffer [24] of 10^6 transitions ($\alpha = 0.6$, $\beta : 0.4 \to 1$, sum-tree implementation) with importance-sampling correction. The target network is synchronized every 10^3 gradient steps. Hidden weight matrices are optimized with Muon [16] (Newton–Schulz orthogonalized momentum), while embeddings, biases, and head layers use AdamW; the base learning rate is 3×10^{-4} with $\gamma = 0.99$, batch size 256, and gradient-norm clipping at 10.

4.4 Asynchronous Training System

Nocturne is CPU-bound: stepping the C++ simulator plus Python-side feature extraction dominates wall-clock time, while the Q-network is small enough that per-environment GPU inference wastes the accelerator. We therefore adopt an asynchronous architecture in the spirit of Sample Factory [22], with three roles:

- **Actors.** N_w worker subprocesses each step N_e environments sequentially ($N_w \times N_e$ environments total; 48 workers $\times$ 2 environments in our default configuration). Observations are written into per-worker shared-memory buffers (zero-copy transport), each worker is pinned to CPU cores, and every environment receives a distinct seed to decorrelate experience.
- **Batched inference (main process).** The main process gathers ready observations across workers, performs a single batched forward pass on the GPU, and scatters actions back through pipes. Inference weights are refreshed from the learner every 16 gradient steps.
- **Learner (background process).** A separate process owns the replay buffer and performs gradient steps. Transition batches arrive through a bounded multiprocessing queue with configurable overload policy (block, drop-newest, or drop-oldest); a train-step scheduler enforces a fixed replay ratio (one gradient step per 512 ingested environment steps after a 5,000-step warm-up), preventing both learner starvation and runaway off-policyness. The learner periodically publishes a compact inference-state snapshot through a latest-value queue.

The same training loop supports four execution modes selected by configuration: `dummy` (single-process, sequential; for debugging), `sync` (lock-step vectorized), `async` (the architecture above), and `ray`, in which actor workers become Ray actors [20] and the system scales across a multi-node cluster with scenario data synchronized from object storage to node-local disk on first access. Off-policy learning makes this decoupling benign: the policy lag introduced by asynchronous weight sync is absorbed by the replay buffer rather than biasing the gradient, in contrast to on-policy architectures that require importance-weighted corrections [8].

5 Experimental Setup

Data. We use driving scenarios from the Waymo Open Motion Dataset [9] converted to Nocturne's format, split into train and held-out test scenarios. The reported checkpoint is evaluated for 1,500 episodes using seed 42; the exact number of unique source scenarios and the train/test split cardinalities were not recorded in the supplied evaluation artifact and are therefore not inferred. Each episode replays one scenario for up to 80 steps at 10 Hz; non-ego road users follow their logged trajectories. Scenarios in which the ego is removed at goal or collision follow the standard Nocturne configuration.

Metrics. We report safety, task, trajectory-quality, and comfort metrics: (i) *goal rate*: fraction of episodes in which the ego reaches its goal position within tolerance; (ii) *collision rate*: fraction of episodes terminating in a collision, decomposed into vehicle–vehicle, vehicle–VRU, and road-edge collisions; (iii) *minimum TTZ statistics*: distribution of the per-episode minimum time-to-zero against vehicles and against vulnerable road users, as a near-miss measure; (iv) *average and final displacement error* (ADE/FDE): mean and endpoint Euclidean distance between the agent trajectory and the logged human trajectory over the episode, measuring how human-like the executed path is; (v) *smoothness*: mean absolute longitudinal jerk $\overline{|\dddot{x}|}$ and steering-rate magnitude, capturing ride comfort and control chattering – a known concern for discrete action spaces; and (vi) *training throughput*: environment steps per second.

Statistical protocol. The currently completed evaluation contains 1,500 held-out episodes from one trained DDQN checkpoint (training seed 42). For binary goal and collision rates we report 95% Wilson score intervals over episodes. For continuous trajectory metrics we report the episode-level mean $\pm$ standard deviation and the number of valid traces. These intervals quantify evaluation-episode uncertainty, not training-seed uncertainty. Consequently, we do not perform between-method significance tests on the present single-run held-out result. PPO and R-MAPPO are compared only on late training-window returns.

Stratified evaluation. Aggregate rates hide where a policy fails. Because every evaluation episode reports a fixed surrounding-agent count of 50, traffic-density curves are not identifiable from the artifact. We instead stratify safety metrics along two recoverable axes: (i) *episode length* (1-step; 2–10; 11–30; 31–50; 51–80 steps), which proxies scenario duration and interaction complexity; and (ii) *vehicle TTZ imminence* (< 1 s, 1–2 s, 2–3 s, ≥ 3 s, or missing), isolating performance in near-conflict regimes.

Algorithm comparison. We compare three RL families trained to $\sim$400k episodes on the same occlusion-aware environment: double dueling DQN, PPO, and risk-aware R-MAPPO (two reported runs). For DDQN we additionally report a deterministic held-out evaluation over 1,500 validation episodes (Table 1). For PPO and R-MAPPO we report the late 1,000-episode training windows at episodes 400k, 420k, and 430k, including mean reward, exploration rate, and terminal goal distance (`goal_dist`).

Training details. The completed DDQN run used synchronous vectorization with four workers and 12 environments per worker (48 parallel environments), a replay buffer of 10^6, and the hyperparameters of Section 4.3; the full configuration is in Appendix A. The contiguous late-training log reaches episode 347,000 with 11,757,888 environment steps and 22,955 learner updates at approximately 283 environment frames per second; additional windows at episodes 400,000 and 405,000 mark the evaluated checkpoint region. PPO and R-MAPPO late windows are reported at episodes 400,000–430,000.

6 Results

6.1 Training Dynamics

The console trace records a clear improvement in the exploration-policy return over the completed run. During episodes 1,000–10,000, the ten reported 1,000-episode window averages have mean -49.69 ± 8.91. By episodes 338,000–347,000, after exploration reached its $\epsilon = 0.05$ floor, the corresponding ten window averages have mean 77.03 ± 9.62, an increase of 126.73 reward units. The highest logged window is 102.21 at episode 332,000, while the final logged

Table 1: Held-out collision-avoidance performance for the evaluated DDQN checkpoint (1,500 episodes). Continuous metrics are episode-level mean ± standard deviation over 1,089 valid multi-step traces. Rate intervals are reported in the text.

Method	Goal (%) ↑	Collision (%) ↓	Min-TTZ p10 (s) ↑	ADE (m) ↓	FDE (m) ↓	Jerk (m/s^3) ↓
DDQN (ep. 400k ckpt)	**95.33**	5.13	0.43	2.24 ± 2.13	6.40 ± 6.20	10.27 ± 4.08

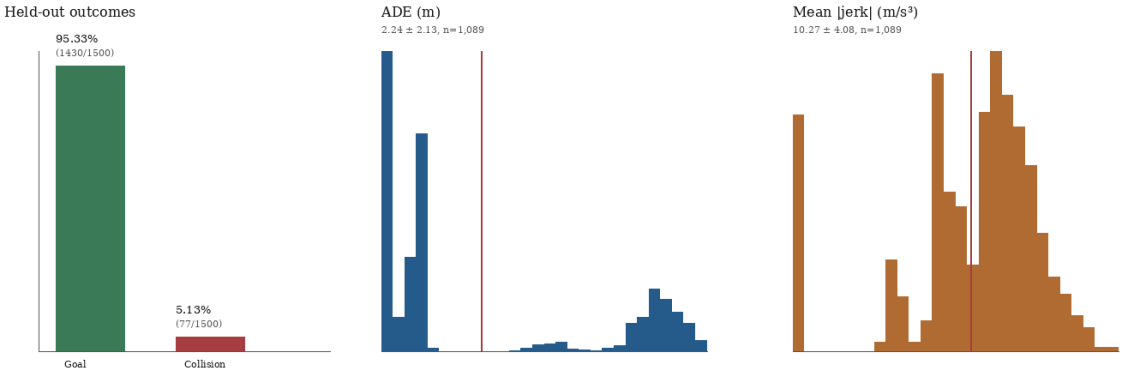


Figure 2: Held-out results for the evaluated DDQN checkpoint. Left: goal and collision event rates across 1,500 episodes; seven episodes contain both event labels. Center and right: episode-level ADE and mean absolute jerk distributions over the 1,089 valid multi-step traces. Dashed lines mark means; missing one-step traces are not imputed.

window is 74.09 at episode 347,000. Across all 17 supplied late-training windows (episodes 331,000–347,000), the mean is 76.23 ± 22.89; the larger spread is driven mainly by the 4.22 window at episode 331,000.

The run also demonstrates stable collection throughput: frames per second increase from 228 at episode 1,000 to 283 in the late-training trace, the replay buffer reaches its 10^6 capacity by episode 36,000, and the final record contains 11,757,888 environment steps and 22,955 learner updates. The printed `q`, `lag`, and `dropped` fields are queueing telemetry rather than Q-value estimates; all remain zero in synchronous mode. These training returns were collected under epsilon-greedy exploration and are therefore not substitutes for deterministic held-out evaluation. The separate 1,500-episode evaluation below provides that test.

6.2 Held-out DDQN Evaluation

The evaluated checkpoint reaches the goal in 1,430 of 1,500 episodes (95.33%; 95% Wilson CI: 94.15–96.29%) and records collisions in 77 (5.13%; 95% Wilson CI: 4.13–6.37%). Seven episodes carry both labels, which reflects the evaluator's event accounting and means the two rates should not be interpreted as complementary outcomes. Mean return is 109.99 ± 48.84, and mean episode length is 20.63 ± 17.85 steps.

Trajectory metrics are available for 1,089 episodes: ADE is 2.24 ± 2.13 m, FDE is 6.40 ± 6.20 m, and mean absolute jerk is 10.27 ± 4.08 m/s^3. Vehicle TTZ has mean 2.04 ± 1.71 s and a 10th percentile of 0.43 s; pedestrian TTZ is available for 362 episodes, with mean 2.40 ± 0.11 s. The omitted trajectory values correspond to 411 one-step episodes for which ADE, FDE, jerk, smoothness, and vehicle TTZ were not emitted. We therefore report the valid sample size and do not impute those observations.

6.3 Algorithm Comparison: DDQN, PPO, and R-MAPPO

Table 2 reports the late-training windows for all three algorithms at episodes 400k–430k. PPO maintains high and stable training reward ($153.53 \rightarrow 153.78 \rightarrow 152.90$) with low `goal_dist` (3.6–7.6 m at 400–420k; 0.6 m at 430k). Both R-MAPPO runs track similarly at 400–420k (run A: 135.53/135.78; run B: 168.60/168.85) but collapse at episode 430k (13.90 and 16.10) with `goal_dist` jumping to 45.78 m—a clear late-training instability. The DDQN occlusion run at

Table 2: Late training windows at episodes 400k–430k. DDQN held-out eval is separate (Table 1).

Method	Episode	Avg reward	ϵ	Goal dist (m)
PPO	400k	153.53	0.154	3.6
	420k	153.78	0.153	7.6
	430k	152.90	0.154	0.6
R-MAPPO (A)	400k	135.53	0.135	5.55
	420k	135.78	0.147	11.90
	430k	13.90	0.190	45.78
R-MAPPO (B)	400k	168.60	0.135	3.6
	420k	168.85	0.147	7.6
	430k	16.10	0.145	45.78
DDQN (train)	400k	55.94	0.050	2.8

Table 3: Held-out DDQN rates stratified by episode length and vehicle TTZ.

Stratum	n	Goal (%)	Collision (%)
Episode length			
1-step	411	100.0	0.0
2–10	2	0.0	100.0
11–30	741	95.8	5.1
31–50	126	70.6	29.4
51–80	220	100.0	0.0
Vehicle TTZ			
< 1 s	412	85.0	15.3
1–2 s	312	97.4	4.5
2–3 s	3	100.0	0.0
≥ 3 s	362	100.0	0.0
missing	411	100.0	0.0

episode 400k logs 55.94 mean reward with `goal_dist`=2.8 m under ϵ=0.05; its deterministic held-out evaluation (Table 1) is substantially stronger (95.33% goal, 5.13% collision), showing that exploration-inflated training returns understate DDQN's deployed performance.

PPO achieves the highest stable training return among the three families, but without a matching zero-ϵ held-out eval we cannot rank safety. R-MAPPO's collapse suggests the Lagrangian safety critic does not stabilize under this training budget. DDQN is the only method with a completed held-out safety evaluation in the dataset.

6.4 Conflict Imminence and Episode Length

Figure 4 and Table 3 show that failures are highly non-uniform. Episodes with vehicle TTZ < 1 s ($n = 412$) have an 85.0% goal rate and 15.3% collision rate, whereas episodes with TTZ ≥ 3 s ($n = 362$) achieve 100% goals and 0% collisions. Length strata tell a complementary story: mid-horizon episodes (31–50 steps; $n = 126$) are hardest (70.6% goal, 29.4% collision), while 1-step episodes ($n = 411$) are all labeled goal successes with missing trajectory metrics, and 51–80-step episodes ($n = 220$) also reach 100% goals. The 2–10-step bin is tiny ($n = 2$) and fully colliding; we report it for completeness but do not over-interpret it. Together these strata support a concrete claim: residual risk concentrates in imminent-conflict regimes rather than being sprinkled uniformly across the test set.

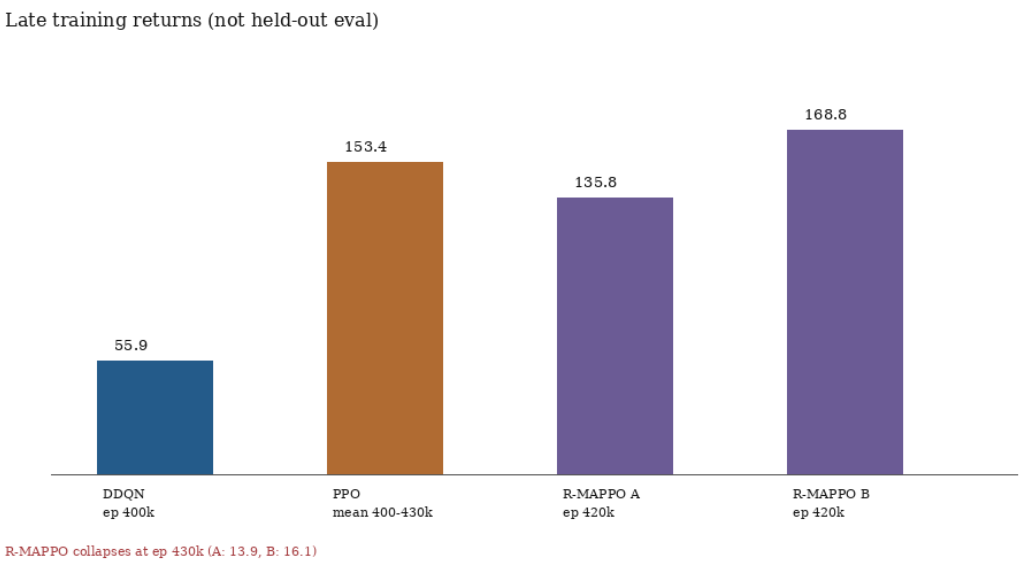


Figure 3: Late training-window mean reward at episode 400k (PPO mean over 400–430k windows). R-MAPPO bars use the 420k pre-collapse windows; both runs fall to $\sim$14–16 reward at 430k.

6.5 Safety–Efficiency Operating Point

With $R_c = 100$ (and the TTZ barrier at $\tau = 3\,\text{s}$, $c_z = 0.5$), the held-out operating point is 95.33% goal rate at 5.13% collision rate. A multi-value penalty sweep was not completed in the supplied artifacts, so we do not draw a Pareto curve. The single measured point already shows that the shaped reward yields high goal completion with a non-zero residual collision rate, consistent with an explicit safety–efficiency compromise rather than a hard constraint.

6.6 Scaling and Throughput

The supplied synchronous DDQN run improves from 228 to 283 environment frames per second in the contiguous late log (439 at the episode-400k window) and reports zero queue lag and dropped transitions. Matched asynchronous or multi-node scaling curves were not supplied.

6.7 Scope of Reported Experiments

Component ablations (noisy layers, PER, n-step, velocity channels, occlusion summaries) and behavior cloning were not included in the final training log bundle. The occlusion sensor-robustness sweep was started but interrupted before completion. Within the provided data, the algorithm comparison covers DDQN (held-out eval + training), PPO (training), and R-MAPPO (two training runs); stratified held-out analysis uses episode length and vehicle TTZ from the DDQN evaluation JSON.

7 Limitations

Non-ego road users replay logged trajectories and do not react to the ego, so the agent cannot learn to exploit (or be robust to) reactive behavior of others; closed-loop multi-agent training is a natural extension.

Discrete vs. continuous control. Our 15-action discrete space (3 throttle $\times$ 5 steering levels) makes value-based learning tractable and exploration well-behaved, but it limits fine maneuvering and can induce control chattering, which we quantify with the jerk and steering-rate metrics of Section 5. Continuous control via actor–critic methods (e.g., SAC

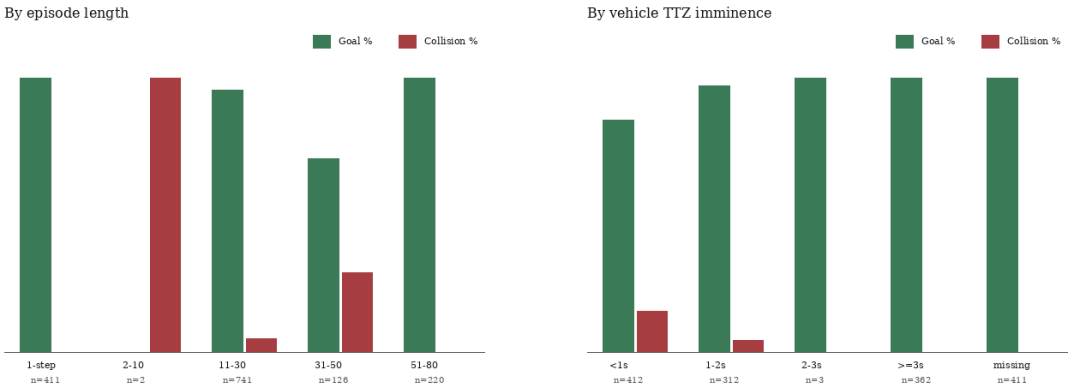


Figure 4: Held-out goal and collision rates stratified by episode length (left) and vehicle TTZ imminence (right). Collision mass concentrates in low-TTZ and mid-horizon episodes.

or TD3) would permit smoother trajectories at the cost of a harder exploration problem in sparse-conflict scenarios; a finer discretization (Nocturne supports up to 630 actions including head-angle control) is an intermediate point. A systematic discrete-vs-continuous comparison under identical observations and rewards is important future work; our smoothness metrics are designed to make that comparison well-posed.

Safety guarantees. Our approach is risk-*aware*, not risk-*guaranteed*: the TTZ barrier shapes behavior away from imminent conflicts but provides no formal constraint satisfaction, and the residual collision rate – 5.13% at the measured R_c=100 operating point – remains above the threshold required for deployment in safety-critical systems. Combining the learned policy with a verified shield or control-barrier filter is a natural direction.

Evaluation scope. The training log includes a three-way comparison at episodes 400k–430k (DDQN, PPO, two R-MAPPO runs). Only DDQN has a matching 1,500-episode held-out evaluation with goal/collision rates and trajectory metrics. The held-out JSON does not vary `num_vehicles` across episodes (all 50), so density stratification is replaced by episode-length and TTZ strata.

Other limitations. The occupancy grid discards agent identity and history beyond relative velocity; our evaluation horizon (8 s) is short relative to full-route driving; and results are in a 2D simulator, so transfer to perception-driven stacks is out of scope.

8 Conclusion

We presented a risk-aware deep RL system for collision avoidance under physically grounded partial observability, trained on real-world driving logs: an observation design that makes occlusion risk explicit through ray-cast visibility filtering and dedicated occlusion-risk summaries, a TTZ-barrier shaped reward that provides a dense safety gradient before conflicts become unavoidable, and a stratified evaluation that localizes residual risk. In the completed single-checkpoint evaluation, the DDQN agent attains a 95.33% goal rate and a 5.13% collision rate across 1,500 held-out episodes, with ADE of 2.24 ± 2.13 m and FDE of 6.40 ± 6.20 m over the 1,089 valid multi-step traces. Collisions concentrate where vehicle TTZ is below one second (15.3% collision rate) and in mid-horizon episodes. Late PPO training returns exceed DDQN exploration returns, while R-MAPPO exhibits a sharp late collapse in both reported runs; only DDQN currently has held-out evaluation. Treating the safety–efficiency compromise as an explicit operating point, together with log-replay realism and high-throughput off-policy training, is a practical recipe for studying safety-critical driving behaviors at scale.

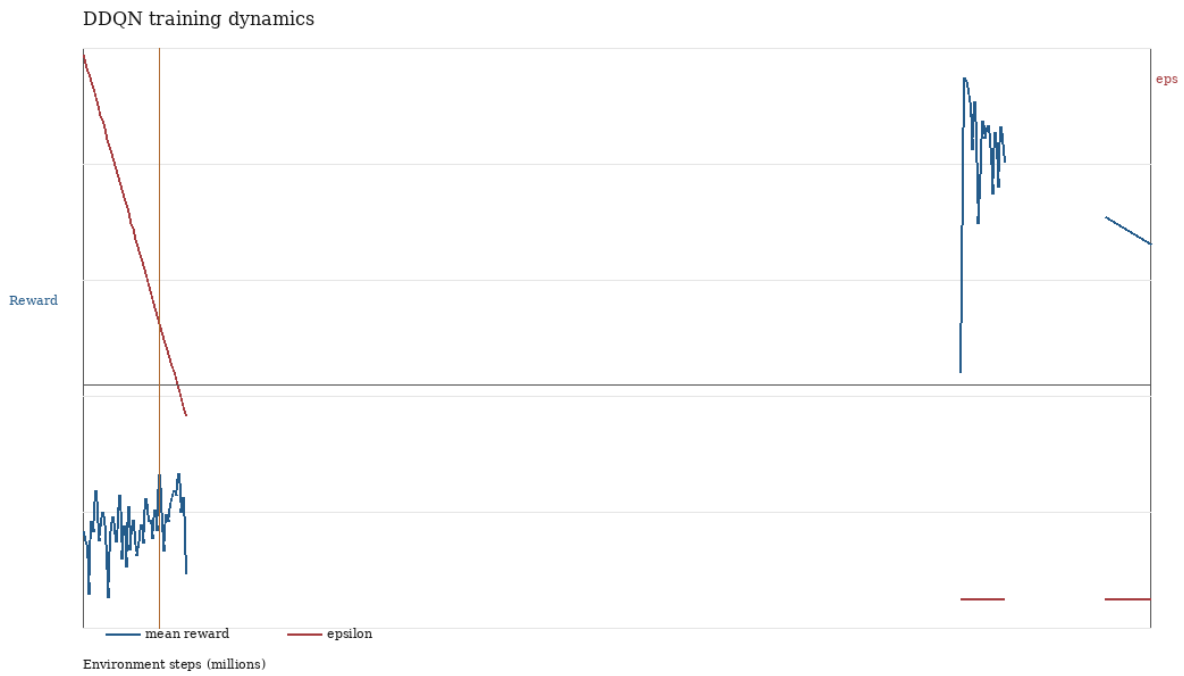


Figure 5: Logged DDQN training dynamics. Each blue point is a reported 1,000-episode mean return and the dashed red curve is the epsilon-greedy exploration schedule. Disconnected segments indicate intervals absent from the supplied transcript and are intentionally not interpolated.

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A Hyperparameters

Table 4: Full training configuration.

Agent		**System**	
Learning rate	3×10^{-4}	Workers N_w	48
Discount γ	0.99	Envs per worker N_e	2
n-step	3	Vec-env mode	async
Batch size	256	Learner queue size	2^{20}
Replay buffer size	10^6	Overload policy	block
Min replay before train	5,000	Weight sync interval	16 steps
Train frequency	1 / 512 env steps	Max train steps per cycle	32
Target update	every 1,000 steps	TF32	on
PER α	0.6	cuDNN benchmark	on
PER β	$0.4 \rightarrow 1.0$	**Reward**	
ϵ decay	$1.0 \rightarrow 0.05$ over $5{\times}10^5$	Goal bonus R_g	100
Hidden layers	4×256	Collision penalty R_c	100
Dueling / Noisy	yes / yes	Off-road penalty R_o	50
Optimizer	Muon + AdamW	Step cost c_s	0.1
Grad-norm clip	10	Progress scale c_p	2.0
Observation		Heading scale c_h	0.3
Grid ($C{\times}H{\times}W$)	$3 \times 20 \times 10$	TTZ threshold τ	3.0 s
Range (fwd/back/lat)	40 / 5 / 10 m	TTZ scale c_z	0.5
VRU / veh / edge weights	2.0 / 1.0 / 3.0	Episode horizon	80 steps
TTZ clip	20 s	Simulation Δt	0.1 s

B Additional Results

For the completed 1,500-episode evaluation, the goal count is 1,430 (95.33%; 95% Wilson CI: 94.15–96.29%) and the collision count is 77 (5.13%; 95% Wilson CI: 4.13–6.37%). Seven episodes have both labels. Continuous statistics are: return 109.99 ± 48.84 ($n = 1{,}500$), episode length 20.63 ± 17.85 steps ($n = 1{,}500$), ADE 2.24 ± 2.13 m ($n = 1{,}089$), FDE 6.40 ± 6.20 m ($n = 1{,}089$), jerk 10.27 ± 4.08 m/s^3 ($n = 1{,}089$), smoothness 10.14 ± 1.13 ($n = 1{,}089$), vehicle TTZ 2.04 ± 1.71 s ($n = 1{,}089$), and pedestrian TTZ 2.40 ± 0.11 s ($n = 362$).

PPO late windows: 153.53 / 153.78 / 152.90 at episodes 400k / 420k / 430k. R-MAPPO run A: 135.53 / 135.78 / 13.90; run B: 168.60 / 168.85 / 16.10 at the same episode indices. DDQN exploration windows at episodes 400k and 405k are 55.94 and 46.94.